

ASHRAY KOTHARI

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EDUCATION

Master of Science - Computer Science

Arizona State University - GPA: 4.0/4.0

Tempe, AZ

August 2023 - May 2025

Bachelor of Technology (Honours) - ICT with a minor in Computational Science

Dhirubhai Ambani Institute of Information and Communication Technology - GPA: 8.8/10.0

Gandhinagar, India

August 2019 - May 2023

SKILLS

- **Programming Languages:** Python, C++, Java, Golang, JavaScript, HTML, CSS
- **Databases:** MySQL, PostgreSQL, MongoDB, DynamoDB, Neptune
- **Frameworks and Libraries:** React.js, Next.js, Express.js, Tailwind, PyTorch, TensorFlow, Flask, FastAPI, OpenMP, JUnit, PyTest
- **DevOps and Cloud Platforms:** AWS, GCP, Docker, Terraform, GitHub Actions, Jenkins, Firebase
- **Software Development:** Git, REST, GraphQL, AJAX, Agile, Microservices, Multithreading, Postman
- **Data Analysis and Visualization:** Tableau, PowerBI, Matplotlib, Seaborn, Numpy, Pandas, Scikit-learn, OpenCV

EXPERIENCE

System Development Engineer Intern | Amazon Inc.

January 2023 - June 2023

- Designed and integrated an automated end-to-end testing framework in the CI/CD pipeline using Jenkins, reducing testing duration by 80% and ensuring rapid, reliable deployments with minimal manual intervention.
- Developed comprehensive monitoring and validation solutions by creating Amazon Quicksight dashboards for system monitoring and expanding a GraphQL-powered regression test suite for Neptune, increasing test coverage by 25%.
- Optimized SQL queries and data pipelines to reduce query execution time by 35% while ensuring 99.9% data accuracy.
- Acted as a primary point of contact among five interns to enhance core frameworks, reducing production bug reports by 40%.

Machine Learning Software Developer | DAIICT

May 2022 - December 2022

- Architected a deep learning model for interferometry-based plasma diagnostics, improving accuracy by 30% and reducing false positives across various conditions and plasma density scenarios.
- Curated and processed a dataset of 5,000+ samples, enhancing model generalization and achieving a near-perfect 0.99 SSIM in plasma density estimation across diverse real-world experimental setups.
- Demonstrated predictive accuracy with low percentage errors (1%-10%), proving the model's ability to handle noisy Erms data and accurately estimate asymmetric plasma profiles in complex diagnostic conditions.
- Co-authored "[Deep learning-assisted microwave-plasma interaction for density estimation](#)", Journal of Physics D: Applied Physics (2024)

PROJECTS

Cloud-based Image Classification System

- Developed a scalable solution that accurately classifies images in real-time while ensuring high availability and fault tolerance under heavy traffic conditions.
- Leveraged AWS EC2 for computing, AWS S3 for secure storage, and employed AWS SQS with multithreading alongside FastAPI, AWS Lambda, and SimpleDB to streamline processing, reduce latency, and enable dynamic resource scaling.

Devconnector

- Built Devconnector, a social network for developers using Next.js and the MERN stack, enabling seamless profile creation, secure authentication, and enhanced collaboration among users.
- Integrated JWT-based security and the GitHub API to dynamically display top repositories, boosting user engagement by 25% and fostering an active developer community.

Pagerank Parallelization

- Achieved a 43x throughput gain and 44x runtime reduction in the PageRank algorithm by parallelizing computations with OpenMP for shared memory architectures, significantly enhancing overall performance.
- Enhanced memory efficiency by designing and implementing a 2D bitset adjacency matrix, which substantially reduced storage overhead in dense graphs while maintaining high computational efficiency during iterative PageRank calculations.
- Researched and identified the optimal data structure for storing node and edge data in PageRank calculations, establishing a 0.062 density cutoff for 1000-node networks based on graph benchmarks.

POSITIONS OF RESPONSIBILITY

Graduate Student Assistant – Grader, Enterprise Learning

January 2024 - Present

- Assisted 100+ students in mastering programming by grading assignments, offering feedback, and mentoring to simplify complex concepts.

Club Lead, AI Club, DA-IICT

August 2021 - December 2022

- Held eight workshops on machine learning and AI, covering topics from neural networks to computer vision with hands-on projects, enhancing skills for 30+ club members.

Event Co-ordinator Lead, Synapse Committee, DA-IICT

August 2020 - December 2022

- Organized university events by managing logistics, sponsorships, and a team of 50+ volunteers, boosting student participation by 30%.

CERTIFICATIONS

- [Solutions Architect Associate SAA-C03](#) (Amazon Web Services - 2024), [Machine Learning Specialization](#) (Coursera - 2022), [Web Development Bootcamp](#) (Udemy - 2020)